

Madhan Kumar S

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EDUCATION

J.J.College of Engineering and Technology

B.Tech in Artificial Intelligence and Data Science - CGPA : 8.02

Trichy

Aug. 2023 – May 2027

Shri Jayendra Vidhyalaya CBSE School

SSLC : 81% - HSC : 63%

Musiri

June 2016 – May 2023

TECHNICAL SKILLS

Languages: Java, SQL

Developer Tools: Git, GitHub, Docker, AWS, IntelliJ

Soft Skills : Problem Solving, Teamwork

EXPERIENCE

Software Development Engineer Intern

Cothon Solutions

Aug. 2025 – Sep. 2025

Hyderabad, Telangana

- Built an AI RAG assistant in Next.js and FastAPI that integrates live stock charts, real-time news lookups, and document vector searches
- Developed FastAPI REST endpoints for news, stocks, web search, and AI chat, integrated with a Supabase PostgreSQL database.
- Secured user authentication with JWTs and migrated all application data storage to Supabase PostgreSQL.

PROJECTS

Autonomous AI Knowledge Worker

Python, TypeScript, Next.js, FastAPI, Supabase, pgvector, Tailwind CSS

- Built an AI RAG assistant combining live stock data, real-time news, and document vector search.
- Engineered a database caching layer tracking 40+ stocks with 15-minute updates and 7-day histories.
- Integrated JWT security, isolated file uploads, and activity log tracking into a responsive dashboard.

AI Grievance System

TypeScript, CSS, HTML, JavaScript, Supabase

- Built a full-stack civic complaint portal with Next.js, TypeScript & Supabase featuring role-based dashboards for citizens, officers, and chiefs.
- Integrated GPS camera, voice-to-text, and AI photo analysis using Web APIs (Canvas, Speech Recognition, Geolocation) to enable smart, accessible complaint filing.
- Developed AI triage engine with automated email notifications, Leaflet.js complaint map, and real-time admin dashboard to streamline grievance resolution workflows.

User Behaviour Analytics

TypeScript, React, HTML, CSS

- Built a real-time security dashboard with React, Flask & WebSockets featuring JWT auth and role-based access control for 3 user roles.
- Developed a 4-signal ML risk pipeline (Isolation Forest + behavioral profiling + velocity checks) to detect anomalous user activity.
- Deployed with Docker Compose (Nginx + Gunicorn) with automated PDF reports, SMTP/Slack alerts, and pytest test coverage reporting.

CERTIFICATIONS

Introduction to Machine Learning - NPTEL

Java - Apollo Computer Education

Cloud Computing Fundamentals - Udemy

Claude 101 - Anthropic